MINIMAL SURFACES IN HYPERBOLIC GEOMETRY

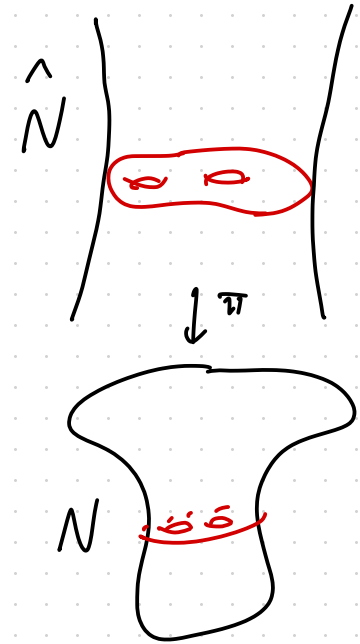
Winter School Côte d'Azur

Lecture II, 6th January 2026

Hyperbolic manifolds

Let (N, h) closed hyperbolic manifold, with $\Gamma \cong \pi_1 S$ a surface subgroup.

Let $\hat{N} \xrightarrow{\pi} N$ the covering such that $\pi_*(\pi_1 \hat{M}) = \Gamma$.



Fact If M is a 3-manifold such that $\pi_1 M \cong \pi_1 S$, then $M \cong S \times \mathbb{R}$.

Consider $[(\hat{N}, \pi^* h)]$

$$AH(S) = \left\{ \begin{array}{l} \text{complete hyperbolic} \\ \text{metrics on } S \times \mathbb{R} \end{array} \right\} / \mathrm{Diffeo}_0(S \times \mathbb{R})$$

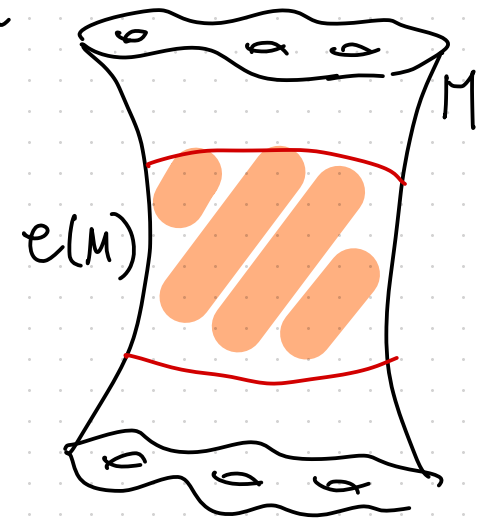
have topologies; geometric/algebraic convergence

The interior of $AH(S)$ is a ball of $\dim_{\mathbb{R}} = 12g-12$

Its elements are quasi-Fuchsian manifolds

<u>Def</u> Let $(\overset{\cong S\times\mathbb{R}}{M}, h)$ be a complete hyperbolic manifold.

M is quasi-Fuchsian if it contains a geodesically convex, compact subset



$\mathcal{C}(M)$ = convex core of M

= smallest such subset

Uhlenbeck's seminal paper (1983)

Recall that

- $B = -\nabla N$ is the shape operator of $\Sigma \subset M$
- the eigenvalues λ, μ of B are the principal curvatures
- Σ is minimal if $\lambda + \mu = 0$

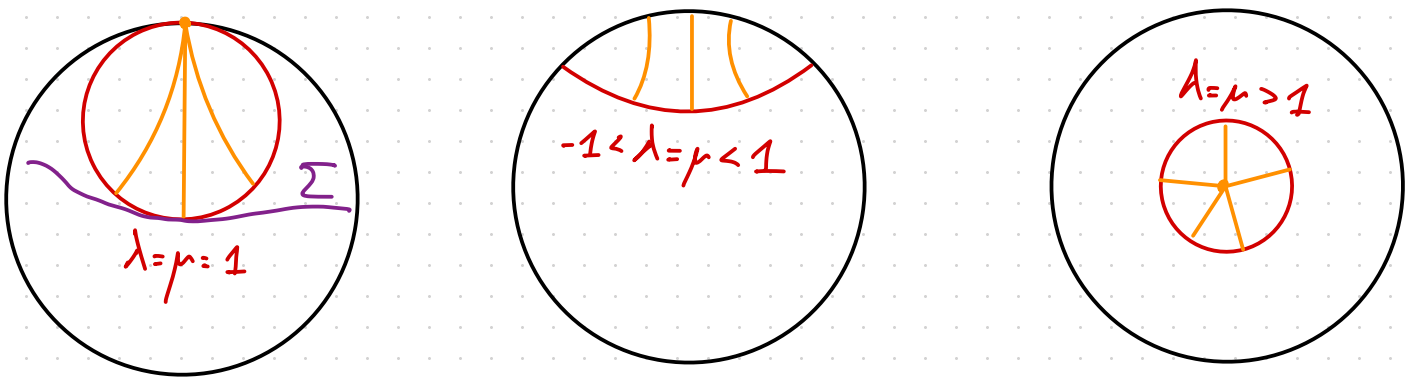
Def Let $(\overset{\cong\, S \times \mathbb{R}}{M}, h)$ be a complete hyperbolic manifold. M is (weakly) almost-Fuchsian if it contains a closed minimal surface such that the principal curvatures are in $(-1,1)$ $([-1,1])$

Thm (Uhlenbeck)

If M is weakly almost-Fuchsian, then it contains a <u>unique</u> minimal surface homotopic to $S \times \{*\}$.

The condition $|\lambda|, |\mu| \leq 1$:

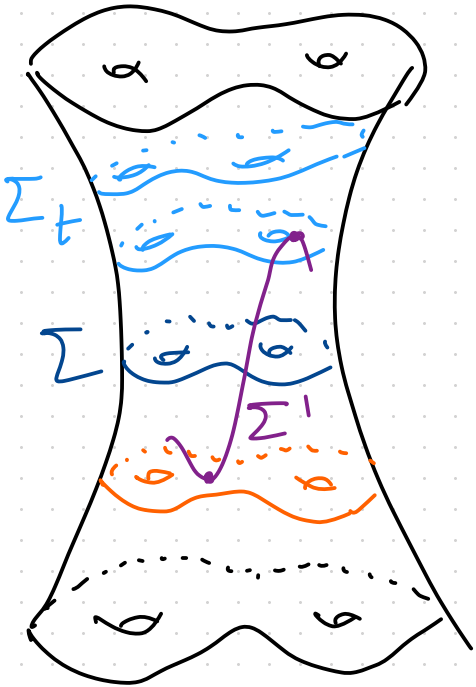
"Σ is less curved than a horosphere"



Sketch of Pf:

The map $\Sigma \times \mathbb{R} \longrightarrow M$ is a diffeo

$$(p, t) \longmapsto \exp_p(tN(p))$$



$t > 0$, $H_{\Sigma_t} < 0$

$t < 0$, $H_{\Sigma_t} > 0$

Geometric maximum principle

$\Downarrow$

uniqueness of Σ

See Sheet 2, Exercises 2 & 3

<u>Summary</u>: WAF $\Longrightarrow$ uniqueness of minimal surface

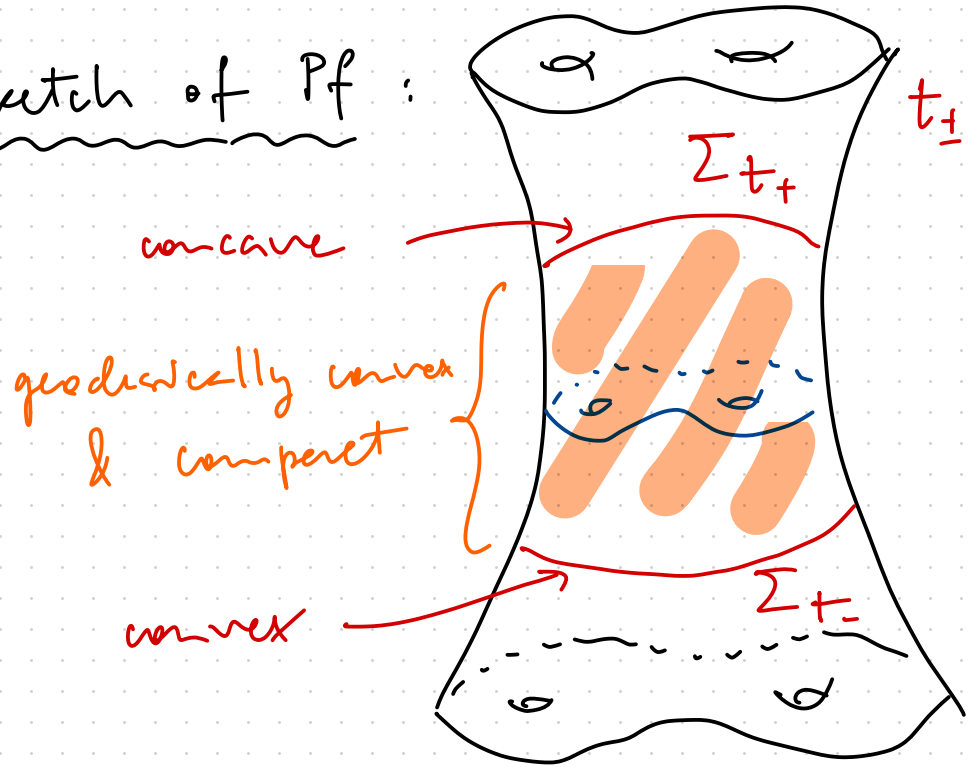
not a necessary condition!

<u>Thm</u> (Huang-Lowe '25)
There exist quasi-Fuchsian manifolds that contain a unique closed minimal surfaces homotopic to $\Sigma \times \{*\}$, that are <u>not</u> weakly almost-Fuchsian.

Thm (Uhlenbeck)

If M is almost-Fuchsian, then it is quasi-Fuchsian.

Sketch of Pf:



$$t_\pm = t_\pm \left(1 - \sup_{p \in \Sigma} \{ |\lambda(p)|, |\mu(p)| \} \right)$$

Rmk: only needs $|\lambda|, |\mu| < 1$, not minimal.

Q: What about WAF?

Theorem (Nguyen - Schlenker - S. '25)

Let (M, h) be a hyperbolic manifold, let $S \subset M$ be an embedded, orientable, two-sided closed minimal surface with principal curvatures in $[-1, 1]$.

Then any neighbourhood U of S in M contains a (non-minimal) surface with principal curvatures in $(-1, 1)$.

Cor If M is weakly almost-Fuchsian, then it is quasi-Fuchsian.

Def Let $(M \cong S \times \mathbb{R}, h)$ be complete hyperbolic.
M is nearly-Fuchsian if it contains a closed surface with principal curvature in $(-1, 1)$

from 2000s

Cor The conjecture "AF $\Leftrightarrow$ NF" is false.

Uhlenbeck openness Davalo '24

$$\mathrm{AF} \overset{\not\Longleftarrow}{\Longrightarrow} \mathrm{WAF} \overset{\not\Longleftarrow}{\Longrightarrow} \mathrm{NF} \overset{\not\Longleftarrow}{\Longrightarrow} \mathrm{QF}$$